

# Can you build a future-proof model from past checkpoints?

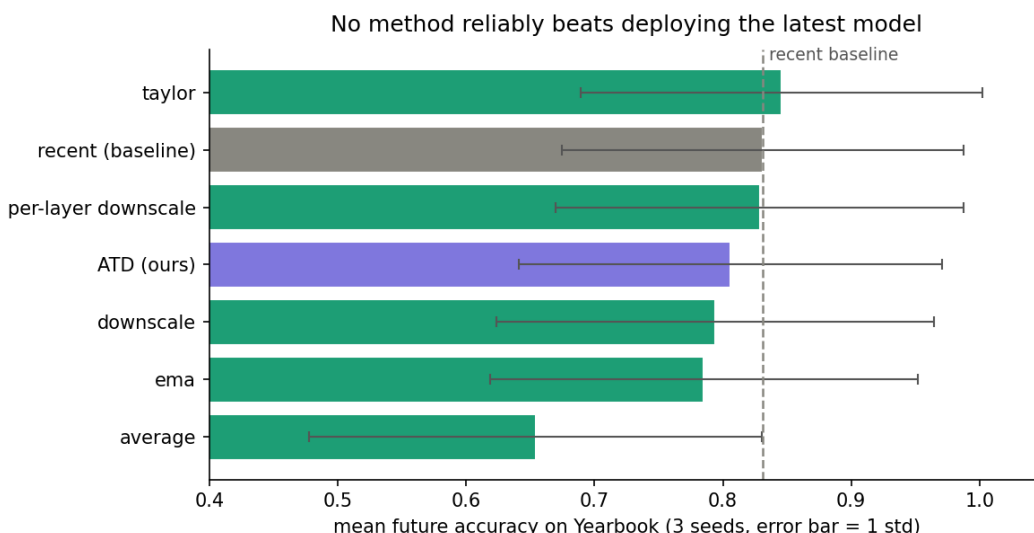
A reproduction and extension of Madaan, Chopra & Cho, “Temporal Generalization: A Reality Check” (ICLR 2026, arXiv:2509.23487). By Omer Ozcan.

## The problem

Models trained on today’s data degrade as the world drifts, and retraining constantly is expensive. So it is tempting to ask: given a history of past model checkpoints, can I compute a model that generalizes to the *future* without ever seeing future data? The paper groups the candidate tricks into interpolation (average or blend past checkpoints, or shrink the latest one) and extrapolation (project the parameter trajectory forward), and evaluates them under a strict rule prior work often breaks: no peeking at the future, including when tuning hyperparameters.

## What I did

I reproduced the core result on the Wild-Time Yearbook benchmark (portraits by year, predict gender) with a small CNN and sequential fine-tuning, across 3 seeds with error bars, tuning every knob on past data only, and I enforced that constraint in code so the tuner cannot index a future timestep. Then I added an original method plus two diagnostics that *measure* the explanations the paper gives for why the methods fail.



## What I found

The central claim reproduced cleanly: no method reliably beats deploying the most recent model. Taylor extrapolation looked like a winner by a hair, but its 1.5-point edge sits inside a 15-point standard deviation, a textbook example of the “apparent improvement that isn’t.” The part I found most interesting: I measured the two mechanisms the paper offers for *why* these methods fail, and on Yearbook neither holds. The loss barriers between consecutive checkpoints are essentially zero, so “disconnected loss basins” does not apply; the basins are connected and the methods still fail. And the correlation the “overconfidence / norm growth” story predicts came out with the wrong sign. The methods fail even though the intuitive reasons do not apply, which strengthens the paper’s deepest point: without assumptions about how the data evolves, the future is arbitrary and no weight-space trick recovers it.

## My original method (ATD)

Anisotropic Trajectory Downscaling: instead of shrinking all parameters toward zero, shrink the model only along the low-dimensional subspace it most recently moved in (the top few directions of recent parameter change), tuned past-only. It sits outside the paper's method family. On Yearbook it did not beat the baseline, an honest negative result, but a useful one: it suggests the vision failure is not a targeted "recent-direction" overfit.

## **Honest scope**

A scaled reproduction on a single vision benchmark with a small model, so a qualitative match rather than exact numbers. The obvious next step is a language track to compare behavior (and basin barriers) across modalities. Code, exact reproduction commands, and all figures:

<https://github.com/OzcanOmer42/temporal-generalization-reality-check>